

Eva Abou Harb

Course: ITAI 1372

Instructor: Adelaja Odujinrin

Midterm Reflection: Cybersecurity Capstone Labs

Date: March 12, 2026

Selection of Labs

For this midterm reflection, I selected three labs that cover different areas of cybersecurity. The first lab is **Pen-testing Fundamentals** from TryHackMe, which focuses on the basic concepts and methodologies used in penetration testing. The second lab is **Intro to Digital Forensics**, also from TryHackMe, which focuses on analyzing digital evidence and investigating cybersecurity incidents. The third lab is **OWASP for LLMs**, which focuses on security risks related to large language models and artificial intelligence systems from the Open Web Application Security Project. These labs cover three distinct areas: penetration testing, digital forensics, and AI security.

Knowledge Gained

From the **Pen-testing Fundamentals** lab on TryHackMe, I learned the basic concepts and process of penetration testing. The first task explained what penetration testing is and how it is used to simulate cyberattacks to identify vulnerabilities in systems before malicious attackers exploit them. The lab also covered the ethical responsibilities involved in penetration testing, emphasizing that security professionals must always obtain permission and follow defined rules of engagement before performing any tests.

Another important concept covered in the lab was penetration testing methodologies, which provide a structured approach for conducting security assessments. I also learned about the differences between black box, white box, and grey box penetration testing. Black box testing simulates an external attacker with no prior knowledge of the system, white box testing provides full system information to the tester, and grey box testing is a combination of both approaches. In the final practical task, the ACME penetration test scenario demonstrated how these concepts can be applied in a realistic environment to identify security weaknesses.

In the **Intro to Digital Forensics** lab, I learned how digital forensics applies computer science to investigate digital evidence for legal or organizational investigations. The lab explained that digital forensics is used in both public-sector investigations, such as law enforcement cases, and private-sector investigations within organizations. I learned that digital evidence can come from many devices, including laptops, smartphones, digital cameras, USB drives, and other storage media.

The lab also introduced the **digital forensics process**, which begins with acquiring the evidence and properly documenting it using a chain of custody to ensure that the evidence has not been tampered with. Investigators must store the evidence securely, transport it to a forensic laboratory, and create a forensic copy so that the original data is preserved. I also learned about the importance of legal authorization, validated forensic tools, and repeatable investigation methods.

In the practical exercise, I explored how investigators analyze document metadata and image EXIF data. Using tools such as `pdfinfo`, I examined metadata from a PDF

document to identify the author of the file. I also learned how ExifTool can extract metadata from images, including GPS coordinates and camera information. Through this process, I discovered the location where a photo was taken and identified the camera model used. This practical example demonstrated how hidden metadata can provide valuable evidence during a digital investigation.

In the **OWASP for LLMs** lab, I explored several vulnerabilities associated with large language models and artificial intelligence systems. These vulnerabilities included prompt injection, sensitive information disclosure, supply chain risks, model and data poisoning, improper output handling, excessive agency, system prompt leakage, vector and embedding weaknesses, misinformation, and unbounded consumption of resources. This lab helped me understand that AI systems can introduce new types of security risks if they are not carefully designed and monitored. Attackers can manipulate AI models through malicious prompts or corrupted training data, which can lead to inaccurate or harmful outputs.

Application in a Professional Setting

The knowledge gained from these labs can be applied in several cybersecurity roles. The skills learned from the **Pen-testing Fundamentals** lab are especially useful for penetration testers and security analysts. By performing controlled security tests, professionals can identify vulnerabilities in systems before attackers exploit them and help organizations strengthen their defenses.

The knowledge gained from the **Intro to Digital Forensics** lab is essential for incident response and investigative roles. When a cyberattack occurs, cybersecurity

professionals must analyze digital evidence to determine how the attack happened, what systems were affected, and what actions the attacker performed. Digital forensics techniques help organizations understand the scope of an incident and prevent similar attacks in the future.

Finally, the knowledge gained from the **OWASP for LLMs** lab is increasingly important as organizations adopt artificial intelligence technologies. Security professionals must understand how AI systems can be manipulated through techniques such as prompt injection or data poisoning. By implementing proper monitoring, input validation, and secure design practices, organizations can reduce the risks associated with AI-powered applications.

Lab References

Pentesting Fundamentals — TryHackMe

<https://tryhackme.com/room/pentestingfundamentals>

Intro to Digital Forensics — TryHackMe

<https://tryhackme.com/room/introdigitalforensics>

OWASP for LLMs — Open Web Application Security Project

<https://application.security/free/llm>